

Lauren Sun

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Research interests: high dimensional statistics, data privacy, information theory, robustness

Education

University of Southern California, Los Angeles, CA
Master of Science, Applied Mathematics
Bachelor of Science, Applied and Computational Mathematics

B.S. GPA: 3.91 | M.S. GPA: 3.78
[Aug 2024 – May 2026](#)
[Aug 2022 – May 2026](#)

Research Projects

- Locally Private Hypothesis Testing with Biased Gaussian Data** | [Master's Thesis](#) | [pdf](#) May 2026
- Chair: Larry Goldstein; Committee Members: Kabir Verchand, Stas Minsker
 - Proved minimax error bounds for binary testing under local differential privacy using LeCam bound, Scheffe's test, concentration inequalities, and Huber-style missing data model
- Differentially Private Median of Means and Trimmed Mean** | [Privacy-Preserving ML Class Project](#) | [pdf](#) Dec 2025
- Propose and implement both private and robust versions of the mean using differentially private quartile subroutine
 - Empirically test algorithms' performance and prod at their level of privacy using membership inference tests
- Organ Sub-Region Segmentation in Whole Slide Images** | [SPIE Medical Imaging](#) | [paper](#) Jun 2024 – Aug 2025
- Implemented and trained a hierarchical CNN model to segment high resolution images into labeled subregions
 - Completed during 2024 Merck internship; oral presentation at SPIE Medical Imaging 2025
- Low-Rank Decomposition for Multi-Party Computation in Machine Learning** | [SCIP Lab](#) Sep 2024 – May 2025
- Implemented neural network decomposition method to improve efficiency of privacy-preserving ML
 - Reproduced training results from other similar papers
 - Awarded \$1,000 USC research stipend to work in Prof. Murali Annavaram's lab
- Decision-Making Under Risk** | [Independent Study](#) Nov 2022 – May 2024
- Led psychology study to quantify and analyze human decision-making under risk using data from Deal or No Deal
 - Programmed Python and JavaScript game played hundreds of study participants
 - Awarded \$3,500 USC research stipend to work with Prof. Richard John
- Generative Machine Learning Models for Data Assimilation** | [Polymath Jr.](#) Jun 2023 – Oct 2023
- Studied optimal transport theory, Bayesian data assimilation, kernels, and other ML methods

Work Experience

- JP Morgan Chase** New York, NY
[Quantitative Research Analyst \(full-time\)](#) incoming Jul 2026
[Quantitative Research Analyst \(intern\)](#) Jun – Aug 2025
- Conduct quantitative analysis on Structured Products pricing data to identify valuation patterns using statistical modeling
 - Develop Python-based tool for trading desks to monitor market dynamics and assist in pricing new deals
 - Communicate with bankers and stakeholders to refine features, validate model effectiveness, and ensure tool usability
- USC Math Department** Los Angeles, CA
[Teaching Assistant](#) Aug 2024 – Feb 2025
- Help grade assignments and exams for Elementary Probability and Statistics (MATH 208x)
 - Hold office hours for 3 hours/week to answer questions and assist students with assignments
- Merck** Rahway, NJ
[Data Science Intern](#) Jun 2024 – Aug 2025
- Developed and implemented computer vision and deep machine learning models to automate analysis of whole slide images
 - Automated the preprocessing of large input images with openslide, PIL, OpenCV, and other libraries
 - Authored research paper and gave oral presentation at SPIE 2025 Medical Imaging conference
- USC Dornsife Office of Admission** Los Angeles, CA
[Student Intern](#) Jan 2023 – May 2024
- Supported outreach/program presentations for prospective students, answered front desk phone calls and emails

On-Campus Involvement

- SCOutfitters – USC Outdoor Club**
[Guide](#) May 2024 – May 2026
[Tech Director](#) May 2025 – May 2026
- Plan and lead hiking, camping, and backpacking trips in groups of 8-10 USC students
 - Complete Wilderness First Responder certification training to prepare for medical emergencies in the backcountry

- Manage ticketing platform, website, and payments through Stripe API

USC Club Tennis

Player Sep 2022 – May 2025

- Compete on A team; successes include 2024 National Finalists, 2025 National 3rd place, 2023 and 2024 Texas Invite Champs

Technical Skills

Programming Languages	Python (numpy, PyTorch, pandas, opencv, Flask, stripe), C++, Java, SQL
Tools	Git, LaTeX

Awards & Recognition

Scholarships

Dean Joan Merit Scholarship \$7,500	2025
Lenore B. Kreiger Endowed Scholarship Fund for Math \$6,000	2025
Dean Joan Merit Scholarship \$7,500	2024
Lenore B. Kreiger Endowed Scholarship Fund for Math \$5,000	2024
Eide Family Endowed Scholarship \$3,000	2023
Presidential Scholarship Half Tuition	8 Semesters
National Merit Scholarship Winner \$3,000	2022

Research Awards

Provost Undergraduate Research Fund (PURF) \$1,000	Spring 2025
Summer Undergraduate Research Fund (SURF) \$3,000	Summer 2023
Student Opportunities for Academic Research (SOAR) \$500	Spring 2023

Certifications

Wilderness First Responder Wilderness Medical Associates International	Jan 2025
Exam P (Probability) Pass Society of Actuaries	Jul 2024
Social and Behavioral Research Best Practices for Clinical Research CITI Program	Feb 2023